

Dimension-saturating local tomography of off-diagonal Fourier-interferometer errors from displaced dark events

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Dark output events are useful null tests for linear-optical devices, but their unprobed probabilities have no signed linear response to a coherent error. We append positive and negative calibrated output displacements and use their count difference as a phase-sensitive local observable. We prove that number-state inputs leave an exact $(m-1)$ -dimensional gauge at an m -mode Fourier interferometer, whereas a phase-calibrated fixed-number multimode cat removes that continuous obstruction. For every $m \geq 2$ and every $n > 2$ divisible by m , an explicit family of dark outcomes and two fixed probes gives a full-rank local Jacobian for all $m(m-1)$ off-diagonal Hermitian error coordinates using four programmed settings. The scalar response count saturates a dimension bound within the declared weak-probe model. We also give exact four-mode certificates, a weighted local estimator, finite-angle and finite-shot tests, and an augmented analysis showing that unknown cat amplitudes as well as phases are confounded with device errors. The construction is an identifiability result, not a scalable replacement for conventional optical characterization: at $n = m$ its reference probabilities decay as $O(\epsilon^2 m^{2-m})$ for unit-spectral-norm probes.

I. INTRODUCTION

Destructive multiphoton interference produces output events with exactly zero ideal probability. Fourier suppression laws generalize the Hong–Ou–Mandel effect to multiport devices [1–5], and forbidden-event populations have become practical witnesses of interference and device quality [2, 6, 7]. A scalar forbidden-event fraction, however, does not specify which coherent device error produced the counts.

There is a more basic local difficulty. If an event is dark at a nominal device, its probability has zero derivative with respect to every coherent error coordinate. The leading quadratic response cannot distinguish opposite errors. The situation parallels the non-identifiability of exact null measurements in quantum statistics. Girotti, Godley, and Guță showed that a deliberate displacement restores identifiability and can support optimal estimation in general pure-state models [8]. Here we develop a finite-mode, photon-counting implementation of that principle: a known signed optical probe supplies a reference amplitude against which unknown leakage interferes.

The central issue is not merely sensitivity but *identifiability*. Number-state preparation and number-resolved detection possess independent input and output phase gauges, familiar from linear-optical characterization [9]. At a Fourier device, the transported input-phase gauge lies entirely inside the nominally off-diagonal output-error coordinates. We prove that number-state experiments cannot see these directions even if an arbitrary calibrated analyzer is appended. An exact four-photon search then saturates, rather than evades, this rank bound.

The missing resource is an input phase reference. We use equal superpositions of all-bunched number states. Fourier multiports and multimode cat states have a substantial prior literature [10]; the associated modular dark sectors also follow from the general symmetry suppression framework [4]. We use this known selection rule to produce a dimension-saturating frame of simultaneously dark outcomes for arbitrary mode number. General near-identity qudit-unitary estimation has also been studied with a controlled-qudit architecture [11]; our construction instead uses bosonic Fourier interference, number-resolved dark outcomes, and two fixed global probes.

Our contributions are:

1. a general count-level signed-probe theorem, including the two-real-row bound per dark event and a rigorous $O(\epsilon^2)$ finite-angle bias bound;
2. an exact $(m-1)$ -dimensional number-state gauge no-go theorem;
3. an exact rank-nine F_4 certificate using one four-photon Fock input;
4. for all $m \geq 2$ and $n > 2$ divisible by m , a dimension-saturating Fourier-cat construction identifying all $m(m-1)$ off-diagonal Hermitian coordinates from $m(m-1)/2$ dark outcomes and four programmed signed settings;
5. a sparse exact F_4 implementation with a rational full-rank certificate, an explicit weighted estimator, and finite-shot simulations.

The result is local tomography on the declared $m(m-1)$ -dimensional off-diagonal slice. It is not global unitary tomography and does not identify the $m-1$ output-diagonal $SU(m)$ generators.

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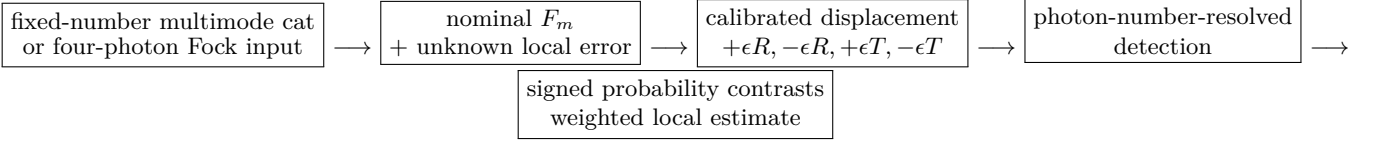


FIG. 1. Protocol concept. The known displacement supplies a reference amplitude at nominally dark outputs. The scheme assumes that the input coefficients and displacement generators have been calibrated.

II. MODEL AND PARAMETER TARGET

Let U_0 be an m -mode passive interferometer and $\hat{U}_0$ its action on a fixed n -photon sector. For number occupations $\mathbf{r}, \mathbf{s}$, the normalized transition amplitude is

$$\mathcal{A}_{\mathbf{r}, \mathbf{s}}(U) = \frac{\text{per } U[\mathbf{s}, \mathbf{r}]}{\sqrt{\prod_j r_j! \prod_k s_k!}}. \quad (1)$$

Rows and columns are repeated according to $\mathbf{s}$ and $\mathbf{r}$.

The unknown coherent output generator belongs to

$$\begin{aligned} \mathcal{H}_{\text{off}}(m) &= \{H = H^\dagger : H_{aa} = 0\}, \\ \dim_{\mathbb{R}} \mathcal{H}_{\text{off}}(m) &= m(m-1). \end{aligned} \quad (2)$$

For the explicit four-mode certificates we use

$$(F_4)_{kj} = \frac{i^{kj}}{2}, \quad 0 \leq k, j < 4. \quad (3)$$

The twelve off-diagonal Hermitian generators are

$$\begin{aligned} X_{pq} &= |p\rangle\langle q| + |q\rangle\langle p|, \\ Y_{pq} &= -i|p\rangle\langle q| + i|q\rangle\langle p|, \quad 0 \leq p < q < 4. \end{aligned} \quad (4)$$

We fix the coordinate order

$$\boldsymbol{\theta} = (x_{01}, y_{01}, x_{02}, y_{02}, x_{03}, y_{03}, x_{12}, y_{12}, x_{13}, y_{13}, x_{23}, y_{23}) \quad (5)$$

and write $G(\boldsymbol{\theta}) = \sum_{\mu} \theta_{\mu} G_{\mu}$. The unknown device is locally

$$U(\boldsymbol{\theta}) = e^{iG(\boldsymbol{\theta})} F_4. \quad (6)$$

The normalization $\text{Tr } X_{pq}^2 = \text{Tr } Y_{pq}^2 = 2$ fixes the meaning of the reported coordinates and condition numbers.

The three traceless output-diagonal generators are deliberately excluded. For an output occupation $\mathbf{s}$, a diagonal output unitary multiplies its amplitude by a phase, leaving its number probability unchanged. Moreover, at a dark event

$$\langle \mathbf{s} | i\hat{Z} \hat{U}_0 | \psi \rangle = iz(\mathbf{s}) \langle \mathbf{s} | \hat{U}_0 | \psi \rangle = 0. \quad (7)$$

They therefore do not occur in the leading dark-event Jacobian. A finite noncommuting analyzer can create higher-order sensitivity to them, but that is a different parameter model.

III. SIGNED-PROBE IDENTIFIABILITY

Consider events $e = (|\psi_e\rangle, |\mathbf{s}_e\rangle)$ that are dark at U_0 :

$$\langle \mathbf{s}_e | \hat{U}_0 | \psi_e \rangle = 0. \quad (8)$$

For a real generator coordinate $\mathbf{u}$, define the complex amplitude functional

$$\ell_e \mathbf{u} = \langle \mathbf{s}_e | i\hat{G}(\mathbf{u}) \hat{U}_0 | \psi_e \rangle. \quad (9)$$

A. Why an undisplaced null is insufficient

Without a probe,

$$p_e(\boldsymbol{\theta}) = |\ell_e \boldsymbol{\theta}|^2 + O(\|\boldsymbol{\theta}\|^3). \quad (10)$$

Thus $Dp_e(0) = 0$. More strongly, no collection of such unprobed dark probabilities has a locally Lipschitz left inverse on an open parameter neighborhood: along $\boldsymbol{\theta} = t\mathbf{u}$, a Lipschitz inverse would imply $|t|\|\mathbf{u}\| = O(t^2)$, a contradiction. This does not exclude a nonregular higher-order inverse on a restricted domain; it rules out the regular local tomography needed for a stable signed estimate.

B. Calibrated central contrast

Append a known probe $H(\mathbf{h})$ after the unknown error and program both signs:

$$p_{e, \mathbf{h}}^{\sigma}(\boldsymbol{\theta}, \epsilon) = \left| \langle \mathbf{s}_e | e^{i\sigma\epsilon\hat{H}(\mathbf{h})} e^{i\hat{G}(\boldsymbol{\theta})} \hat{U}_0 | \psi_e \rangle \right|^2, \quad \sigma = \pm 1. \quad (11)$$

We use the baseline-subtracted response

$$\begin{aligned} R_{e, \mathbf{h}}(\boldsymbol{\theta}, \epsilon) &= \epsilon^{-1} [p_{e, \mathbf{h}}^+(\boldsymbol{\theta}, \epsilon) - p_{e, \mathbf{h}}^-(\boldsymbol{\theta}, \epsilon) \\ &\quad - p_{e, \mathbf{h}}^+(0, \epsilon) + p_{e, \mathbf{h}}^-(0, \epsilon)]. \end{aligned} \quad (12)$$

This is four times the convention $[p^+ - p^-]/(4\epsilon)$ often used for a central difference.

Theorem 1 (signed-probe Jacobian). For every finite-dimensional photon sector,

$$\left. \frac{\partial}{\partial \theta_{\mu}} \lim_{\epsilon \rightarrow 0} R_{e, \mathbf{h}}(\boldsymbol{\theta}, \epsilon) \right|_{\boldsymbol{\theta}=0} = 4 \text{Re}[(\ell_e \mathbf{h})^* (\ell_e e_{\mu})]. \quad (13)$$

At finite angle the derivative differs from Eq. (13) by $O(\epsilon^2)$.

A proof and a norm-explicit remainder bound are given in Appendix A. Equation (13) is a statement about observable probability differences, not an assumption that complex transition amplitudes can be measured directly. The probe amplitude $\ell_e \mathbf{h}$ acts as a phase reference.

Corollary 1 (two-row event bound). For one dark event, all limiting signed-probe rows lie in

$$\text{span}_{\mathbb{R}}\{\text{Re } \ell_e, \text{Im } \ell_e\}. \quad (14)$$

Hence one event contributes at most two independent real first-order measurements, no matter how many probes are applied.

If a limiting $q \times d$ response matrix has full column rank, some $d \times d$ row minor is nonsingular. Continuity preserves that minor for all sufficiently small nonzero calibrated ϵ . The ordinary local inverse-function theorem applied to the selected d responses then gives a local differentiable left inverse. This argument must not be confused with the algebraic Jacobian conjecture, which concerns a polynomial self-map with constant nonzero determinant and a *global polynomial* inverse. Nor does full rank at one point imply global injectivity. Our map need not be polynomial, the determinant is evaluated locally, and only local identifiability is claimed.

IV. THE NUMBER-STATE GAUGE AND ITS OPTIMAL RANK

Theorem 2 (transported input-phase gauge). Let F_m be the m -mode Fourier matrix. For every real traceless diagonal D , the generator

$$K_D = F_m D F_m^\dagger \quad (15)$$

has zero diagonal and is therefore an off-diagonal Hermitian generator. Nevertheless, for every input number state $|\mathbf{r}\rangle$,

$$e^{i\hat{K}_D} \hat{F}_m |\mathbf{r}\rangle = \hat{F}_m e^{i\hat{D}} |\mathbf{r}\rangle = e^{i\sum_j d_j r_j} \hat{F}_m |\mathbf{r}\rangle. \quad (16)$$

Consequently K_D is invisible to all output number probabilities, even after an arbitrary known output analyzer. The invisible subspace has dimension $m - 1$.

The proof is included in Appendix B. At $m = 4$, an explicit basis inside the coordinate order (5) is

$$\begin{aligned} K_A &= X_{01} + X_{03} + X_{12} + X_{23}, \\ K_B &= Y_{01} - Y_{03} + Y_{12} + Y_{23}, \\ K_C &= X_{02} + X_{13}. \end{aligned} \quad (17)$$

Thus Fock-input experiments can identify at most nine of the twelve off-diagonal coordinates.

A. Exact four-photon saturation

The bound is tight. Use input $\mathbf{r} = (1, 1, 1, 1)$ and the nine output-probe contrasts in Table I. All five outputs

TABLE I. An exact rank-nine design. Multiple probes on one output give separate scalar contrasts.

output $\mathbf{s}$	probes
(0, 0, 1, 3)	X_{03}, Y_{03}
(0, 0, 2, 2)	X_{02}, Y_{02}
(0, 0, 3, 1)	X_{12}, Y_{12}
(0, 1, 1, 2)	X_{23}, Y_{23}
(0, 2, 2, 0)	Y_{02}

are exactly dark by the cyclic Fourier selection rule [1, 4].

The normalized 9×12 limiting matrix is displayed and derived in Appendix B. It annihilates exactly the three vectors in Eq. (17); a 9×9 minor has determinant $81/8$. Therefore its rank is nine. Nine scalar contrasts are necessary for a nine-dimensional quotient, and five events are necessary by Corollary 1. The design attains both lower bounds.

V. A COHERENT PHASE REFERENCE

A. Known Fourier-cat dark sectors

Let

$$F_m(k, j) = m^{-1/2} \omega^{kj}, \quad \omega = e^{2\pi i/m}, \quad (18)$$

and define the phase-twisted all-bunched cat, equivalently a fixed-number generalized multimode NOON state,

$$|\Psi_\ell^{(m,n)}\rangle = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \omega^{-\ell j} |n e_j\rangle. \quad (19)$$

For an output occupation $\mathbf{s}$ of total size n , put $Q(\mathbf{s}) = \sum_k k s_k \pmod{m}$. Direct evaluation gives

$$\mathcal{A}_{\Psi_\ell, \mathbf{s}}(F_m) = \sqrt{\frac{n!}{\prod_k s_k!}} m^{(1-n)/2} \delta_{Q(\mathbf{s}), \ell}. \quad (20)$$

Thus all outputs outside one modular sector are dark. The Fourier multiport-cat connection is known from Vourdas and Dunningham [10], and Eq. (20) is also an immediate cyclic-symmetry selection rule [4]. We recall it to specify our phase convention and resource; we do not claim it as a new suppression law. A self-contained derivation is given in Appendix C.

B. An all-mode dimension-saturating frame

We now specialize to the charge-zero cat, take $n > 2$ divisible by m , and write

$$\alpha_{\mathbf{s}} = \sqrt{\frac{n!}{\prod_a s_a!}} m^{(1-n)/2}. \quad (21)$$

For an output with $Q(\mathbf{s}) = c \neq 0$, define the phase-free dark-amplitude functional

$$L_{\mathbf{s}}(H) = \alpha_{\mathbf{s}} \sum_{a=0}^{m-1} s_a H_{a,a-c}, \quad (22)$$

where indices are modulo m . Its physical amplitude derivative is $\ell_{\mathbf{s}}(H) = iL_{\mathbf{s}}(H)$.

Theorem 3 (all-mode Fourier-cat frame). For every $m \geq 2$ and every $n > 2$ divisible by m , there are $m(m-1)/2$ dark outcomes and two fixed Hermitian probes R, T for which the two limiting signed contrasts per outcome have full rank on $\mathcal{H}_{\text{off}}(m)$. Thus four settings, $\pm\epsilon R$ and $\pm\epsilon T$, suffice for local identification at all sufficiently small calibrated nonzero ϵ .

Here is the explicit construction. Choose the cyclic separations $c = 1, \dots, \lfloor m/2 \rfloor$. If $2c \not\equiv 0 \pmod{m}$, use all m outcomes

$$\mathbf{s}_{p,c} = (n-1)\mathbf{e}_p + \mathbf{e}_{p+c}, \quad 0 \leq p < m. \quad (23)$$

If m is even and $c = m/2$, retain only $0 \leq p < m/2$. Their charge is $np + c \equiv c$, so all are dark. Their total number is $m(m-1)/2$.

Let R be the real complete-graph probe,

$$R_{ab} = 1 \ (a \neq b), \quad R_{aa} = 0. \quad (24)$$

Orient every edge of cyclic separation $c < m/2$ by

$$T_{a,a-c} = i, \quad T_{a-c,a} = -i. \quad (25)$$

For even m , orient the remaining antipodal edges as $T_{p,p+m/2} = i$ for $0 \leq p < m/2$, with the reverse entries fixed by Hermiticity. Then

$$\begin{aligned} L_{p,c}(R) &= \alpha n, & L_{p,c}(T) &= i\alpha n & (2c \not\equiv 0), \\ L_{p,c}(R) &= \alpha n, & L_{p,c}(T) &= i\alpha(n-2) & (2c \equiv 0), \end{aligned} \quad (26)$$

where the selected outcomes share $\alpha = \sqrt{n} m^{(1-n)/2}$. The two probes therefore read the real and imaginary parts of every selected complex functional.

Both probes are lossless mode-mixing Hamiltonians and can in principle be synthesized from pairwise beam splitters and phases, but they are dense: $O(m^2)$ couplings must be compiled into each analyzer. Since $\|R\|_2 = m-1$ and $\|T\|_2 \leq m-1$, a conservative unit-spectral-norm implementation is $\bar{R} = R/(m-1)$ and $\bar{T} = T/(m-1)$; reference amplitudes and response rows acquire the same known scale factors. With the unnormalized generators the perturbative condition is $\epsilon(m-1) \ll 1$, whereas for $\bar{R}, \bar{T}$ it is $\epsilon \ll 1$.

For completeness, the underlying block inversion is especially simple. For a non-antipodal separation, set $z_a = H_{a,a-c}$. Equation (22) becomes

$$\frac{L_{p,c}(H)}{\alpha} = (n-1)z_p + z_{p+c}. \quad (27)$$

The map is $((n-1)I + P_c)z$ with P_c a permutation. A vector in its kernel would obey $(n-1)\|z\| = \|z\|$, and

hence vanishes because $n > 2$. For the antipodal class, with $z_p = x_p + iy_p$,

$$\frac{L_{p,c}(H)}{\alpha} = (n-1)z_p + z_p^* = nx_p + i(n-2)y_p, \quad (28)$$

which is also invertible. The cyclic edge classes partition all off-diagonal Hermitian entries, proving the theorem. A detailed derivation is in Appendix C.

One dark event supplies at most two real rows by Corollary 1. Consequently the theorem's $m(m-1)$ scalar contrasts attain the dimension lower bound, and its selected outcomes attain the induced event-count bound, *within this weak-probe regular-local framework*. These are algebraic counts, not experimental run counts: all bins are collected in parallel, and an overcomplete outcome set may be statistically preferable. The operationally notable resource is four programmed settings independent of m . This does not prove global injectivity, nor does it prove that four settings are necessary among more general protocols. After division by the known reference magnitudes in Eq. (26), each cyclic block has condition number at most $n/(n-2)$. Without that row calibration, the worst antipodal gain ratio is at most $[n/(n-2)]^2$.

For $m \geq 3$, the smallest member of the family is $n = m$; for $m = 2$ the first allowed choice is $n = 4$. At $n = m$, the common squared amplitude normalization is

$$\alpha^2 = m^{2-m}. \quad (29)$$

This exponential-in- m factor is a probability-rate cost, not a rank or conditioning failure. With the dense unnormalized probes above, the probe-only dark-event probability scales as $p_{\text{ref}} = O(\epsilon^2 m^{4-m})$ at $n = m$; if the probe operators are normalized to fixed spectral size it scales instead as $O(\epsilon^2 m^{2-m})$. These estimates caution against interpreting the algebraic all-mode theorem as an immediate large- m sample-complexity advantage.

C. A sparse exact F_4 implementation

Set $m = n = 4$, $\ell = 0$:

$$|\Psi_{\text{cat}}\rangle = \frac{|4000\rangle + |0400\rangle + |0040\rangle + |0004\rangle}{2}. \quad (30)$$

Equation (20) makes the 25 outputs with $Q(\mathbf{s}) \neq 0$ exactly dark. Use the two probe generators

$$H_X = X_{01} - X_{02}, \quad H_Y = Y_{01} - Y_{02}. \quad (31)$$

They are the two quadratures coupling mode 0 to the antisymmetric supermode $(|1\rangle - |2\rangle)/\sqrt{2}$.

In the column orders

$$\begin{aligned} \boldsymbol{\theta}_X &= (x_{01}, x_{02}, x_{03}, x_{12}, x_{13}, x_{23}), \\ \boldsymbol{\theta}_Y &= (y_{01}, y_{02}, y_{03}, y_{12}, y_{13}, y_{23}), \end{aligned} \quad (32)$$

six H_X contrasts and six H_Y contrasts have the limiting matrices

$$J_X = \begin{pmatrix} 0 & -\frac{3}{2} & 0 & 0 & -\frac{3}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{3}{4} & 0 & \frac{3}{2} \\ \frac{9}{4} & 0 & 0 & \frac{3}{4} & 0 & 0 \\ \frac{3}{4} & 0 & \frac{3}{4} & 0 & 0 & \frac{3}{2} \\ 0 & -4 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & \frac{3}{2} & \frac{3}{4} & 0 & 0 \end{pmatrix}, \quad (33)$$

and

$$J_Y = \begin{pmatrix} 0 & -\frac{3}{2} & 0 & 0 & -\frac{3}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{3}{4} & 0 & \frac{3}{2} \\ 0 & -\frac{3}{2} & 0 & 0 & \frac{3}{2} & 0 \\ \frac{9}{4} & 0 & 0 & \frac{3}{4} & 0 & 0 \\ \frac{3}{4} & 0 & -\frac{3}{4} & 0 & 0 & \frac{3}{2} \\ \frac{3}{4} & 0 & -\frac{3}{2} & \frac{3}{4} & 0 & 0 \end{pmatrix}. \quad (34)$$

The H_X output order is

$$(0022), (0112), (0310), (1021), (1030), (1102), \quad (35)$$

and the H_Y output order is

$$(0022), (0112), (0220), (0310), (1021), (1102). \quad (36)$$

Every omitted cross block is exactly zero. The exact determinants are

$$\det J_X = -\frac{243}{8}, \quad \det J_Y = \frac{729}{32},$$

$$\det \text{diag}(J_X, J_Y) = -\frac{177147}{256}. \quad (37)$$

The Gaussian-integer derivation is in Appendix C.

This sparse four-mode certificate is an implementation alternative to Theorem 3. It uses seven distinct outcomes rather than the six in the general construction, while retaining twelve scalar responses and four programmed settings because all outcomes for a probe are recorded in parallel: $\pm\epsilon H_X$ and $\pm\epsilon H_Y$. The raw limiting condition number is 7.05; row normalization gives 4.86, and ideal background-free Poisson whitening gives 4.79. These are local design diagnostics, not hardware-independent resource guarantees. Here row normalization divides every response row by its Euclidean norm, while Poisson whitening divides a probability-derivative row by the square root of its nominal displaced-event probability before singular values are taken.

VI. FINITE PROBE ANGLE AND COUNT STATISTICS

The matrices (33)–(34) are the $\epsilon \rightarrow 0$ Jacobian of R , not exact finite-angle Jacobians. For the cat input, exact finite-angle probabilities are available from

$$A_s(U) = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \prod_{k=0}^3 U_{kj}^{s_k}, \quad p_s(U) = |A_s(U)|^2. \quad (38)$$

TABLE II. Finite-angle Jacobian bias and spectral condition number in the ideal coherent model. Bias is relative Frobenius error from the limiting matrix.

ϵ	relative bias	$\kappa_2(J_\epsilon)$
0.010	7.74×10^{-4}	7.043
0.020	3.09×10^{-3}	7.033
0.050	1.92×10^{-2}	6.968
0.100	7.46×10^{-2}	6.759

For the probes (31), $H^3 = 2H$, so

$$e^{i\epsilon H} = I + \frac{i \sin(\sqrt{2}\epsilon)}{\sqrt{2}} H + \frac{\cos(\sqrt{2}\epsilon) - 1}{2} H^2. \quad (39)$$

Appendix A proves that the central-contrast Jacobian has no $O(\epsilon)$ error: its first correction is $O(\epsilon^2)$. For the selected design, exact finite-angle evaluation gives the compact diagnostic in Table II. The favorable condition number at larger angle does not remove the growing model bias. Probe-only baseline subtraction in Eq. (12) is also essential: for the selected H_Y outcomes, the largest undivided baseline begins as $9\epsilon^3/8 + O(\epsilon^5)$.

For independent selected Poisson channels,

$$Y_{as} \sim \text{Pois}\{N_a[p_{as}(\boldsymbol{\theta}) + b_s]\}, \quad (40)$$

the Fisher information is

$$I_{\mu\nu}^P = \sum_{a,s} N_a \frac{\partial_\mu p_{as} \partial_\nu p_{as}}{p_{as} + b_s}. \quad (41)$$

At a displaced dark event, $p = c\epsilon^2 + O(\epsilon^3)$ and $\partial_\mu p = d_\mu \epsilon + O(\epsilon^2)$. The information therefore has a finite nonzero limit when $b = 0$, but is suppressed as $\epsilon^2/(c\epsilon^2 + b)$ above a fixed background floor. A multinomial model must additionally include the negative covariance between outcomes recorded in one setting. Under any stated likelihood, a Cramér–Rao bound is conditional on local unbiasedness and does not by itself give a hardware cost forecast.

A. Estimator, allocation, and all-mode scaling

For a calibrated finite angle, let $\hat{\mathbf{r}}$ collect the empirical positive-minus-negative probability differences, divided by ϵ , after subtracting the probe-only differences. In the statistical model used below, these baselines are pre-calibrated and treated as exact; their calibration shots are not included in N . If instead they are estimated from N_0 finite-count trials, their covariance must be added to Σ , N_0 must enter the resource budget, and calibration drift produces a systematic offset. Thus “four settings” describes the tomography stage, conditional on an offline baseline and probe calibration. If J_ϵ is the exact finite-angle Jacobian and Σ is the count covariance, a concrete

local estimator is weighted least squares,

$$\hat{\theta} = (J_\epsilon^\top \Sigma^{-1} J_\epsilon)^{-1} J_\epsilon^\top \Sigma^{-1} \hat{r}. \quad (42)$$

It supplies the local covariance approximation $(J_\epsilon^\top \Sigma^{-1} J_\epsilon)^{-1}$ and an ellipsoidal confidence region, conditional on the calibrated model. A finite-angle maximum-likelihood fit to the probabilities in Eq. (38) is preferable once the initial estimate leaves the linear regime.

The all-mode construction permits an explicit ideal allocation calculation. Use the unit-spectral-norm probes $\bar{R}, \bar{T}$, let N be the total number of successful input trials, and split each probe-pair allocation equally between its two signs. For a non-antipodal cyclic block define

$$C_c = \sum_{k=0}^{m-1} |n-1 + \exp(2\pi i k/m)|^{-2},$$

$$C = \sum_{c=1}^{\lfloor (m-1)/2 \rfloor} C_c. \quad (43)$$

The trace coefficients for the R and T pairs are

$$(A_R, A_T) = \begin{cases} (C, C), & m \text{ odd}, \\ \left(C + \frac{m}{2n^2}, C + \frac{m}{2(n-2)^2}\right), & m \text{ even}. \end{cases} \quad (44)$$

Minimizing the trace Cramér–Rao bound gives

$$w_R = \frac{\sqrt{A_R}}{\sqrt{A_R} + \sqrt{A_T}}, \quad w_T = 1 - w_R,$$

$$\min \text{Tr Cov}(\hat{\theta}) = \frac{(\sqrt{A_R} + \sqrt{A_T})^2}{4N\alpha^2}. \quad (45)$$

This result is background-free and local. At $n = m$, its average per-coordinate variance scales as

$$\frac{1}{m(m-1)} \text{Tr Cov}(\hat{\theta}) = \Theta\left(\frac{m^{m-4}}{N}\right). \quad (46)$$

Moreover, the weakest nominal reference probability is $\Theta(\epsilon^2 m^{2-m})$. Obtaining even a constant expected number of reference counts in every sign bin therefore requires

$$N = \Omega(\epsilon^{-2} m^{m-2}). \quad (47)$$

If additive backgrounds obey $b_s = \Theta(b)$ uniformly over the selected bins and are stable across the two signs, and if this fixed b dominates the signal in those bins, the average local bound instead scales as

$$\Theta\left(\frac{b m^{2m-6}}{N \epsilon^2}\right). \quad (48)$$

Thus neither row normalization nor algebraic dimension saturation implies sample efficiency.

As an end-to-end check, we simulated the sparse F_4 protocol with all twelve coordinates nonzero, $\|\theta\|_2 =$

TABLE III. All-mode resource diagnostic for $n = m$, unit-spectral-norm probes, and $\epsilon = 0.05$. The last two columns are respectively N times the ideal average Cramér–Rao variance and the total trials needed for ten expected events in the weakest sign bin under the A-optimal allocation.

m	α^2	$p_{\min}$	$N\bar{\text{Var}}$	N_{10}
4	6.25×10^{-2}	6.94×10^{-5}	1.09	5.17×10^5
6	7.72×10^{-4}	1.23×10^{-6}	27.2	3.11×10^7
8	3.81×10^{-6}	7.01×10^{-9}	2.75×10^3	5.59×10^9
10	1.00×10^{-8}	1.98×10^{-11}	6.27×10^5	2.00×10^{12}

0.002, $\epsilon = 0.05$, $N = 2 \times 10^6$, and calibrated background $b = 10^{-5}$. Numerical A-optimal allocation for the signed-contrast estimator assigns $0.51635N$ trials to the H_X sign pair. Across 5000 Poisson repetitions, Eq. (42) gives empirical root-mean-square vector error 2.9033×10^{-3} , compared with 2.9002×10^{-3} predicted by its covariance; the nominal 95% ellipsoid covers 95.4% of trials. This validates the estimator in the stated idealized model, not against unmodeled hardware errors. A deterministic scan over 128 random simultaneous-error directions finds worst relative linear-inversion bias 0.9%, 3.4%, and 9.1% on spheres of radius 10^{-3} , 3×10^{-3} , and 10^{-2} , respectively. This is a direction-sampled F_4 diagnostic, not a uniform radius theorem.

Figure 2 adds two compact sweeps. The finite-shot error follows the predicted $N^{-1/2}$ law, while increasing $\|\theta\|_2$ eventually exposes the bias of local linear inversion. Counts from a fixed-trial experiment are multinomial rather than independent Poisson. Here the total nominal probability of the six selected bins is at most 0.0062 per setting at $\epsilon = 0.05$; the omitted multinomial covariance terms are therefore a sub-percent correction to the selected-channel covariance. We use Poissonization to isolate the local estimator and state this approximation explicitly rather than claiming an exact fixed-trial likelihood.

This section remains an ideal-model statistical layer. Loss, partial distinguishability, source contamination, detector crosstalk, and calibration drift require an augmented likelihood before experimental sample complexity can be claimed.

B. Cat-coefficient SPAM confounding

The resource that removes the number-state gauge is the complete complex coefficient vector, not phase coherence alone. Write a perturbed fixed-number multimode cat as

$$|\Psi_\eta\rangle = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} (1 + \eta_j) |n e_j\rangle, \quad (49)$$

where normalization and global phase remove two irrelevant first-order real directions. For a nominally dark

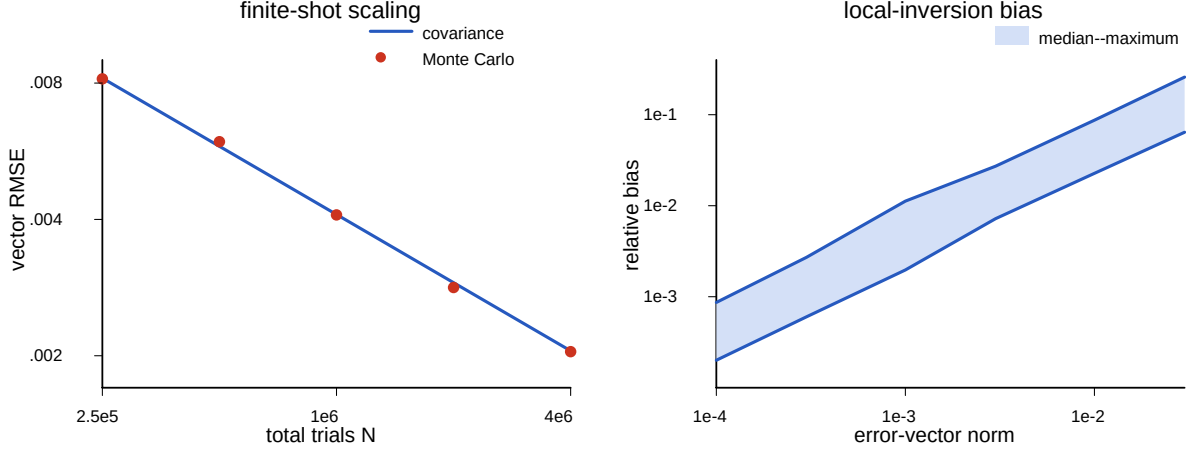


FIG. 2. Sparse- F_4 validation in the calibrated Poisson model. Left: empirical vector RMSE (points) and the covariance prediction (line) versus total tomography trials at $\epsilon = 0.05$ and $\|\theta\|_2 = 0.002$; their common scaling is $N^{-1/2}$. Right: deterministic relative local-inversion bias over 128 sampled directions; the band spans the median to maximum. The figure is generated by the released standard-library script.

charge- c output, direct substitution in the cat polynomial gives

$$\delta A_s = \alpha_s \hat{\eta}_c, \quad \hat{\eta}_c = \frac{1}{m} \sum_{j=0}^{m-1} \eta_j \omega^{jc}. \quad (50)$$

The leakage is constant in p over the selected outcomes $s_{p,c}$.

Proposition 1 (coefficient-device confounding). Every first-order leakage vector in Eq. (50), restricted to the outcomes of Theorem 3, is exactly representable by an off-diagonal device error. For a non-antipodal block, choose all z_p equal so that

$$i[(n-1)z_p + z_{p+c}] = in z_p = \hat{\eta}_c. \quad (51)$$

For the antipodal block, the two real variables in $i[nx_p + i(n-2)y_p]$ reproduce an arbitrary complex constant because $n > 2$. Thus adding unknown cat coefficients to the selected-data Jacobian does not increase its rank; it creates exact device-SPAM null directions.

For real amplitude imbalances with zero mean, the nuisance image has dimension $m-1$. Unknown relative phases have the same dimension. Their combined image has dimension $m-1$ for odd m and m for even m : the two images coincide on non-antipodal Fourier components, while amplitude and phase supply the two distinct quadratures of the antipodal component. In particular, at $m=4$ the amplitude-only and phase-only images each have rank three, but their union has rank four rather than six. The accompanying exact Gaussian-integer test certifies these three ranks.

Phase confounding also has the direct operator interpretation

$$F_m e^{iD} = e^{iF_m D F_m^\dagger} F_m. \quad (52)$$

Complete dephasing therefore returns the experiment to the number-state gauge. The full complex cat coefficient vector must be independently calibrated, or additional inputs and outcomes must be introduced and shown to give a full augmented SPAM-device Jacobian.

Even after that calibration, the construction is not globally injective: for $D = \text{diag}(d_0, \dots, d_{m-1})$, the discrete conditions

$$n(d_j - d_0) \in 2\pi\mathbb{Z} \quad (1 \leq j < m) \quad (53)$$

leave the equal-phase cat unchanged up to a global phase. The corresponding finite transported transformations $F_m \exp(iD) F_m^\dagger$ are exact global aliases. They are discrete near a calibrated representative and do not contradict local full rank.

VII. STANDARD PROPAGATION OF INTERNAL COHERENT ERRORS

Let a nominal circuit factor as

$$U_0 = U_L U_{L-1} \cdots U_1. \quad (54)$$

If a small coherent generator H_ℓ is inserted after layer ℓ , set $W_\ell = U_L \cdots U_{\ell+1}$. Then

$$U_L \cdots U_{\ell+1} e^{i\delta_\ell H_\ell} U_\ell \cdots U_1 = e^{i\delta_\ell W_\ell H_\ell W_\ell^\dagger} U_0. \quad (55)$$

For simultaneous small errors, the first-order output-equivalent generator is

$$G_{\text{eff}} = \sum_\ell \delta_\ell W_\ell H_\ell W_\ell^\dagger + O(\|\delta\|^2). \quad (56)$$

Thus the signed-probe reconstruction applies to arbitrary internal coherent, lossless errors after adjoint propagation.

Equation (56) is the standard first-order conjugation rule, included to define what the reconstructed output generator means for a layered circuit; it is not claimed as an independent novelty.

This does not automatically locate a faulty component. Different layer errors can have the same output-equivalent generator, and output-diagonal projections remain outside our $m(m-1)$ -coordinate target. Layer localization requires the circuit Jacobian across additional configurations, addressable controls, or structural priors. Appendix E gives the short proof and the multi-error expansion.

VIII. RELATION TO PRIOR WORK AND LIMITATIONS

The deliberate displacement of a null measurement is a general estimation principle, not one introduced here. Girotti *et al.* established its role in resolving null-measurement non-identifiability and in asymptotically optimal pure-state estimation [8]. Our contribution is an explicit multiphoton-count implementation, the exact number-state gauge theorem, and explicit dimension-saturating local frames for a Fourier device.

Likewise, Eq. (20) is a known Fourier-cat/cyclic selection effect [4, 10]. The new role assigned to the cat is operational: it supplies the missing relative phase reference. Standard Fourier suppression experiments [1–3] and programmable photonic processors [6, 12] provide the experimental context, but preparing a phase-stable multimode multiphoton cat is substantially harder than preparing a number state. The rank-nine F_4 Fock design is the nearer-term protocol; the cat family is a complete-identifiability benchmark.

As optical-device characterization, the proper comparator set is much broader than near-identity qudit estimation. One- and two-photon schemes reconstruct arbitrary linear-optical transformations and have been analyzed under loss, noise, and mode mismatch [13, 14]. Coherent-light protocols recover transfer-matrix moduli and phases using a standard laser and intensity measurements [9]; PhaseLift variants are noise robust and efficiently scalable [15]. More recent procedures explicitly model unbalanced input/output losses, phase instability, photon indistinguishability, and experimental reconstruction [16, 17]. These mature methods are the default choice for broad transfer-matrix characterization.

The present protocol should therefore not be read as a lower-cost replacement. Its plausible niche is a *residual, operating-point diagnostic* after conventional characterization: it uses the same multiphoton and photon-counting layer as a Fourier-suppression experiment, keeps the number of analyzer programs fixed, and turns structured forbidden events into signed alarms for coherent

drift. Whether this is preferable to periodic classical-light recalibration is platform dependent and is not demonstrated here.

This distinction also explains why suppression can be scalable for a scalar benchmark while failing here as a high-dimensional reconstruction resource. Recent Fourier protocols aggregate many forbidden outcomes to estimate one indistinguishability parameter with favorable scaling [7]. Our frame instead resolves phase-sensitive information across individual rare bins; summing them would discard the coordinate information. The two uses of suppression therefore have different statistical bottlenecks.

The general high-dimensional unitary-estimation protocol of Escandón-Monardes *et al.* remains a useful conceptual comparator: it reconstructs all $d^2 - 1$ parameters and can attain the quantum Cramér–Rao bound, but uses a target qudit, two controls, and controlled shift and phase operations [11]. To our knowledge, prior work has not combined Fourier-cat suppression with displaced-null readout to give the explicit all-mode local frame proved here. This is a scoped structural claim, not a claim to the first or most efficient linear-optical characterization protocol.

The rigorous scope is:

- local reconstruction around a calibrated nominal interferometer, not global injectivity or global unitary tomography;
- $m(m-1)$ off-diagonal coordinates only, not the $m-1$ output-diagonal $SU(m)$ directions;
- coherent lossless errors with calibrated probes and the complete complex input coefficient vector;
- two probe directions and four signed settings suffice, but neither a one-probe nor a fewer-setting impossibility theorem has been proved;
- the scalar-contrast and induced outcome counts are minimal only within the declared weak-probe regular-local framework; outcome bins are collected in parallel and this algebraic minimality is not a run-cost advantage;
- identification is local and calibrated: global aliases and off-diagonal device errors confounded with uncalibrated input amplitudes, phases, or probe errors are not excluded;
- at $n = m$, $\alpha^2 = m^{2-m}$, so event rates decay rapidly even though the row-normalized condition bound is at most $m/(m-2)$;
- the ideal likelihood does not absorb loss or distinguishability into a universal “background” parameter [18, 19].

TABLE IV. Operational comparison. Configuration counts are schematic because the cited protocols differ in parallel detection and calibration assumptions. “Four settings” does not include preparation calibration.

Protocol	Input complexity	Programmed configurations	Parameters identified	Scope	Noise treatment
Coherent-light reconstruction [9, 15, 16]	Low: standard laser and intensity detectors	Scales with modes	Complex transfer matrix up to declared gauges/loss model	Full matrix; not restricted to a small neighborhood	Experimentally demonstrated; noise, loss, and phase instability treated
One-/two-photon reconstruction [13, 14, 17]	Moderate: single photons or photon pairs	Typically $O(m^2)$ port combinations	Linear-optical transfer matrix	Full matrix; not restricted to a small neighborhood	Loss, mode mismatch, indistinguishability, and uncertainty studied
F_4 Fock dark-event design	Four-photon state	Eight probe directions (sixteen signed settings) in Table I	Nine of twelve off-diagonal coordinates at F_4	Local	Finite-angle ideal model; no complete SPAM/loss model
Fourier-cat dark-event design	Very high: calibrated fixed-number multi-mode cat	Four signed probe settings independent of m	$m(m-1)$ off-diagonal coordinates at F_m	Local	Ideal estimator and count model; exact coefficient-SPAM confounding

IX. DISCUSSION

A multiphoton dark event is not by itself a signed sensor. The known probe in Eq. (12) supplies a reference amplitude and turns photon counts into linear local observables. Which coordinates can then be reconstructed depends on both the collection of dark amplitude gradients and the phase reference carried by the input state.

This viewpoint exposes a sharp resource separation. Number states have an exact $(m-1)$ -dimensional blind subspace at F_m ; the F_4 certificate saturates the resulting rank-nine bound. A calibrated cat state lifts that continuous gauge, and the cyclic-edge construction gives an invertible $m(m-1)$ -coordinate design for every mode number. Unknown cat amplitudes or phases create device-indistinguishable leakage on the selected data. The distinction is therefore one of calibrated identifiability, not simply better visibility.

The most immediate extensions are a joint SPAM-device likelihood, lower-cost coherent reference states, circuit-specific internal-error dictionaries, and sparse or bounded-norm probes that improve large- m count rates. Any such extension must preserve the separation between local full rank, finite-angle bias, finite-shot efficiency, and global uniqueness.

X. CODE AND DATA AVAILABILITY

The exact certificates, finite-angle calculation, resource-scaling analysis, weighted estimator, Monte Carlo simulation, tests, and reproduction instructions are available at <https://github.com/HainanZhao/maph/tree/fourier-dark-tomography-v1.1/projects/fourier-dark-tomography>. The release tag `fourier-dark-tomography-v1.1` fixes the submitted state and resolves to its exact com-

mit; the code is MIT licensed. In particular, `certify_general_fourier_cat_tomography.py` reproduces the all-mode ranks and coefficient-SPAM certificate, `search_su4_dark_tomography.py` reproduces the exact F_4 matrices and determinants, `analyze_cat_finite_statistics.py` reproduces the finite-angle table, and the resource and reconstruction scripts reproduce Table III and the Monte Carlo values quoted above.

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Appendix A: Proof of the signed-probe and finite-angle theorems

Put $|\phi_e\rangle = \hat{U}_0|\psi_e\rangle$ and suppress the event subscript. For a fixed probe $\hat{h}$, define

$$\begin{aligned}
 a(t) &= \langle \mathbf{s} | e^{it\hat{h}} | \phi \rangle, \\
 b_\mu(t) &= \langle \mathbf{s} | e^{it\hat{h}} i\hat{G}_\mu | \phi \rangle, \\
 g_\mu(t) &= \partial_{\theta_\mu} |\langle \mathbf{s} | e^{it\hat{h}} e^{i\hat{G}(\theta)} | \phi \rangle|^2 \Big|_{\theta=0} \\
 &= 2 \operatorname{Re}\{a(t)^* b_\mu(t)\}.
 \end{aligned} \tag{A1}$$

Darkness gives $a(0) = 0$. Let

$$a_r = \langle \mathbf{s} | (i\hat{h})^r | \phi \rangle, \quad b_{r\mu} = \langle \mathbf{s} | (i\hat{h})^r i\hat{G}_\mu | \phi \rangle. \tag{A2}$$

The derivative of the response (12) is

$$\partial_{\theta_\mu} R|_0 = \frac{g_\mu(\epsilon) - g_\mu(-\epsilon)}{\epsilon}. \quad (\text{A3})$$

Taylor expansion of a and b gives

$$\begin{aligned} \partial_{\theta_\mu} R|_0 &= 4 \operatorname{Re}(a_1^* b_{0\mu}) \\ &+ 4\epsilon^2 \operatorname{Re} \left\{ \frac{a_1^* b_{2\mu}}{2} + \frac{a_2^* b_{1\mu}}{2} + \frac{a_3^* b_{0\mu}}{6} \right\} + O(\epsilon^4). \end{aligned} \quad (\text{A4})$$

Since $a_1 = \ell \mathbf{h}$ and $b_{0\mu} = \ell \mathbf{e}_\mu$, Eq. (13) follows. Writing $\ell = \mathbf{u} + i\mathbf{v}$ as a complex row, its probe row is a real linear combination of $\mathbf{u}$ and $\mathbf{v}$, which proves Eq. (14).

For a norm bound, let

$$M = \|\widehat{h}\| \leq n\|h\|, \quad L_\mu = \|\widehat{G}_\mu\| \leq n\|G_\mu\|. \quad (\text{A5})$$

Leibniz's rule applied to Eq. (A1) yields

$$|g_\mu'''(t)| \leq 2 \sum_{k=0}^3 \binom{3}{k} M^k M^{3-k} L_\mu = 16M^3 L_\mu. \quad (\text{A6})$$

The central-difference remainder theorem therefore gives, for the convention (12),

$$|\partial_{\theta_\mu} R|_0 - 4 \operatorname{Re}(a_1^* b_{0\mu})| \leq \frac{16}{3} \epsilon^2 M^3 L_\mu. \quad (\text{A7})$$

This bound is conservative but uniform. Retaining the explicit ϵ^2 term in Eq. (A4) leaves an $O(\epsilon^4)$ remainder. Baseline subtraction does not change the derivative with respect to $\boldsymbol{\theta}$, but it removes the finite-angle probe-only offset from the observed response.

Appendix B: Fock gauge proof and exact rank-nine certificate

For $D = \operatorname{diag}(d_0, \dots, d_{m-1})$,

$$(F_m D F_m^\dagger)_{pp} = \frac{1}{m} \sum_j d_j = \frac{\operatorname{Tr} D}{m} = 0. \quad (\text{B1})$$

Conjugation commutes with exponentiation, so $e^{iK_D} F_m = F_m e^{iD}$ at the single-particle level and hence in every symmetric photon sector. A number state is an eigenvector of $\widehat{D} = \sum_j d_j \widehat{n}_j$, proving Eq. (16). Appending a known unitary V preserves the global phase, so no subsequent number measurement removes the obstruction. The map from traceless D to K_D is injective and its dimension is $m-1$.

For completeness, the F_4 finite certificate can be evaluated with Gaussian integers. Let

$$Z_{\mathbf{r}, \mathbf{t}} = \operatorname{per}(2F_4)[\mathbf{t}, \mathbf{r}], \quad D_{\mathbf{r}, \mathbf{s}} = 2^n \sqrt{\prod_j r_j! \prod_k s_k!}. \quad (\text{B2})$$

For a dark target, the first amplitude derivatives are

$$\begin{aligned} \ell X_{pq} &= \frac{i}{D_{\mathbf{r}, \mathbf{s}}} (s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q} + s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q}), \\ \ell Y_{pq} &= \frac{1}{D_{\mathbf{r}, \mathbf{s}}} (s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q} - s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q}). \end{aligned} \quad (\text{B3})$$

Negative occupations are omitted. These follow directly from the Fock ladder factors; the square roots cancel against the changed-state normalization.

Substitution into Eq. (13), for the row order of Table I, gives

$$J_F = \begin{pmatrix} 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & 0 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & -\frac{3}{2} \\ -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \quad (\text{B4})$$

Each row annihilates the coordinate vectors of K_A, K_B, K_C . The minor in columns

$$(x_{01}, y_{01}, x_{02}, y_{02}, x_{03}, y_{03}, x_{12}, y_{12}, y_{13}) \quad (\text{B5})$$

has determinant $81/8$. Hence $\operatorname{rank} J_F = 9$ and the three independent gauge vectors form its complete nullspace.

Appendix C: Fourier-cat frame proof and sparse rank-twelve certificate

For the component $|n\mathbf{e}_j\rangle$, every term in the repeated-column permanent has the same phase. Equation (1)

gives

$$\mathcal{A}_{|ne_j\rangle, \mathbf{s}}(F_m) = \sqrt{\frac{n!}{\prod_k s_k!}} m^{-n/2} \omega^{jQ(\mathbf{s})}. \quad (\text{C1})$$

Multiplying by the coefficients in Eq. (19) and summing over j gives

$$\frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \omega^{j[Q(\mathbf{s})-\ell]} = \sqrt{m} \delta_{Q(\mathbf{s}), \ell}, \quad (\text{C2})$$

which proves Eq. (20).

We next derive Eq. (22) and complete the all-mode proof. For a general single-particle matrix U , the cat amplitude is the polynomial

$$A_{\mathbf{s}}(U) = \sqrt{\frac{n!}{\prod_a s_a!}} \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \prod_{a=0}^{m-1} U_{aj}^{s_a}. \quad (\text{C3})$$

Along $U(\theta) = e^{i\theta H} F_m$, $\dot{U}_{aj} = i \sum_b H_{ab} F_m(b, j)$. Differentiating Eq. (C3) at zero produces terms proportional to

$$\sum_{j=0}^{m-1} \omega^{j[Q(\mathbf{s})+b-a]} = m \delta_{b, a-c} \quad (\text{C4})$$

when $Q(\mathbf{s}) = c$. Collecting the factorial and Fourier normalizations gives

$$\left. \frac{d}{d\theta} A_{\mathbf{s}}(e^{i\theta H} F_m) \right|_{\theta=0} = i\alpha_{\mathbf{s}} \sum_a s_a H_{a, a-c} = iL_{\mathbf{s}}(H), \quad (\text{C5})$$

as asserted.

For the selected outcome (23), all factorials are the same and $\alpha = \sqrt{n} m^{(1-n)/2}$. In a non-antipodal cyclic class, the m complex coordinates $z_a = H_{a, a-c}$ are independent, because their Hermitian conjugates belong to the unselected class $-c$. The selected functionals form the complex matrix $(n-1)I + P_c$. If it annihilates z , unitarity of the permutation gives

$$(n-1)\|z\| = \|P_c z\| = \|z\|, \quad (\text{C6})$$

so $z = 0$ for $n > 2$. In the self-conjugate antipodal class, Hermiticity instead reduces each edge to $z_p = x_p + iy_p$, and its functional is $nx_p + i(n-2)y_p$. This is invertible over the reals. Each undirected off-diagonal edge appears in exactly one selected cyclic class, so these blocks are a direct-sum decomposition of $\mathcal{H}_{\text{off}}(m)$.

It remains to show that the complex functionals are observable from counts. For the probes defined in Eq. (25), direct substitution gives Eq. (26). Since $\ell = iL$, Theorem 1 reduces here to

$$DR_{\mathbf{s}, h}(0)[H] = 4 \text{Re}\{L_{\mathbf{s}}(h)^* L_{\mathbf{s}}(H)\}. \quad (\text{C7})$$

The R and T rows are therefore nonzero multiples of the real and imaginary parts of every $L_{\mathbf{s}}$. The preceding

direct-sum argument proves full rank. There are exactly $m(m-1)$ such real rows. Since Corollary 1 bounds one event by two rows, the event and scalar-response counts meet the weak-probe regular-local lower bounds. Analytic dependence on ϵ and persistence of a nonzero determinant then establish the finite-angle local statement through the ordinary inverse-function theorem.

The conditioning claim follows from the singular values $|n-1+\lambda|$, $|\lambda| = 1$, of each non-antipodal block, and from the two gains $n, n-2$ in an antipodal block. Their ratio is at most $n/(n-2)$ after known reference-row normalization. Finally, $\alpha^2 = nm^{1-n}$, which becomes Eq. (29) when $n = m$.

We now certify the sparse F_4 matrices without numerical zero tests. For any four-mode unitary,

$$A_{\mathbf{s}}(U) = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \prod_{k=0}^3 U_{kj}^{s_k}. \quad (\text{C8})$$

Its derivative in the direction $\dot{U}_{\mu} = iG_{\mu}F_4$ is

$$\dot{A}_{\mathbf{s}, \mu} = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \sum_{k: s_k > 0} s_k (\dot{U}_{\mu})_{kj} U_{kj}^{s_k-1} \prod_{\substack{r=0 \\ r \neq k}}^3 U_{rj}^{s_r}. \quad (\text{C9})$$

At $U = F_4$, all entries lie in $\mathbb{Q}(i)$ times the displayed normalization. The limiting physical response is

$$J_{\mathbf{s}, \mu} = 4 \text{Re} \left(\dot{A}_{\mathbf{s}, h}^* \dot{A}_{\mathbf{s}, \mu} \right). \quad (\text{C10})$$

Evaluating Eq. (C9) for the outputs (35)–(36) gives Eqs. (33)–(34). Direct expansion gives

$$\det J_X = -243/8, \quad \det J_Y = 729/32. \quad (\text{C11})$$

The H_X rows have zero entries in all six Y columns and the H_Y rows have zero entries in all six X columns. Thus the full determinant is their product, Eq. (37). This is an exact rational certificate. The released verification independently constructs the same rows from Gaussian-integer permanent moments and asserts every cross-block zero.

Appendix D: Derivation of the all-mode resource scaling

At the nominal device, one selected block and one probe quadrature have, to leading order,

$$p_{\pm} = \alpha^2 \epsilon^2 |h|^2, \quad \partial_{\theta} p_{\pm} = \pm 2\alpha^2 \epsilon |h| B, \quad (\text{D1})$$

after choosing the probe phase so that the real row B is read. If N_h trials are assigned to that probe pair and split equally between its signs, the signed contrast has Jacobian $J_h = 4\alpha^2 |h| B$ and covariance $4\alpha^2 |h|^2 / N_h$ in

the background-free local model. Its Fisher information is consequently

$$I_h = 4N_h\alpha^2 B^\top B. \quad (\text{D2})$$

The reference magnitude and ϵ cancel only in the background-free Fisher ratio; the factor α^2 does not.

For a non-antipodal separation, $B = (n-1)I + P_c$ for each of the two real quadratures. Therefore

$$\text{Tr}[(B^\top B)^{-1}] = \sum_{k=0}^{m-1} |n-1 + \exp(2\pi i k/m)|^{-2} = C_c. \quad (\text{D3})$$

The antipodal real and imaginary blocks have $m/2$ coordinates and gains n and $n-2$, respectively. Summing the inverse-Gram traces gives Eq. (44). For allocations $N_R = w_R N$, $N_T = w_T N$, the covariance trace is

$$\frac{1}{4N\alpha^2} \left(\frac{A_R}{w_R} + \frac{A_T}{w_T} \right). \quad (\text{D4})$$

Elementary differentiation under $w_R + w_T = 1$ proves Eq. (45).

At $n = m$, every C_c lies between m/n^2 and $m/(n-2)^2$. There are $\Theta(m)$ non-antipodal separations, each contributing $\Theta(1/m)$, so $C = \Theta(1)$. Combining $\alpha^2 = m^{2-m}$

with $m(m-1)$ coordinates proves Eq. (46). For the normalized probes, the weakest reference probability is a constant-in- m gain times $\epsilon^2\alpha^2$, proving Eq. (47). When $b_s = \Theta(b)$ uniformly on the selected bins and $b \gg p_\pm$, Eq. (41) instead contributes $O(N\epsilon^2\alpha^4 m^2/b)$ information per coordinate. Its reciprocal gives Eq. (48). These are local asymptotics in the selected-channel Poisson model and do not include the probability of successfully preparing or transmitting the n -photon input.

Appendix E: Internal-error propagation

Write $U_0 = W_\ell V_\ell$, with $W_\ell = U_L \cdots U_{\ell+1}$ and $V_\ell = U_\ell \cdots U_1$. Then

$$\begin{aligned} W_\ell e^{i\delta H_\ell} V_\ell &= W_\ell e^{i\delta H_\ell} W_\ell^\dagger U_0 \\ &= e^{i\delta W_\ell H_\ell W_\ell^\dagger} U_0, \end{aligned} \quad (\text{E1})$$

which proves Eq. (55) exactly for one inserted error. For several errors, expanding each exponential to first order and collecting terms gives Eq. (56). Commutators enter at second order, so the layer-to-output dictionary is linear only in the local regime.

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